ALTERNATIVE ARCHITECTURES

CastIron Media — Phase 1 Alternate Approaches

This document presents the two architectures evaluated alongside the recommended Balanced Hybrid approach, priced and diagrammed on the same basis, for when cost, operational simplicity, or raw performance should outweigh the recommendation.

1. Purpose and Relationship to the Recommended Approach

The companion document, CastIron Media Phase 1 Technical Approach, recommends Approach 01 (Balanced Hybrid Lakehouse) and covers it in full — requirements, architecture, performance, the clean-room design, delivery, and cost. This document covers the two architectures that were evaluated against it and set aside: Approach 02 (Cost-Optimized Classic Micro-Batch) and Approach 03 (Fully Serverless, Performance-Optimized). All three share the same Phase 1 requirements, the same clean-room pattern, and the same cost methodology — only the compute strategy for streaming, identity resolution, and SQL serving differs between them.

> **Key notes**
>
> **Clean room:** identical across all three approaches — CastIron's own Snowflake provider account, same AWS region as the partner, read-only Iceberg offering linked into the partner's Collaboration Data Clean Room. See the Technical Approach document, Section 6, for the full design; it is not repeated per approach here beyond the summary table in each section below.
>
> **Costs:** approximate list prices (AWS us-east-1, Databricks Premium tier) for planning only, same basis as the Technical Approach document's Section 9 assumptions.

2. Approach 02: Cost-Optimized Classic Micro-Batch

All compute runs on classic clusters in the customer VPC without Photon, using spot workers and scheduled micro-batches instead of continuous streaming. Events are ingested from Firehose's S3 delivery, and the identity graph stays hosted by the licensor.

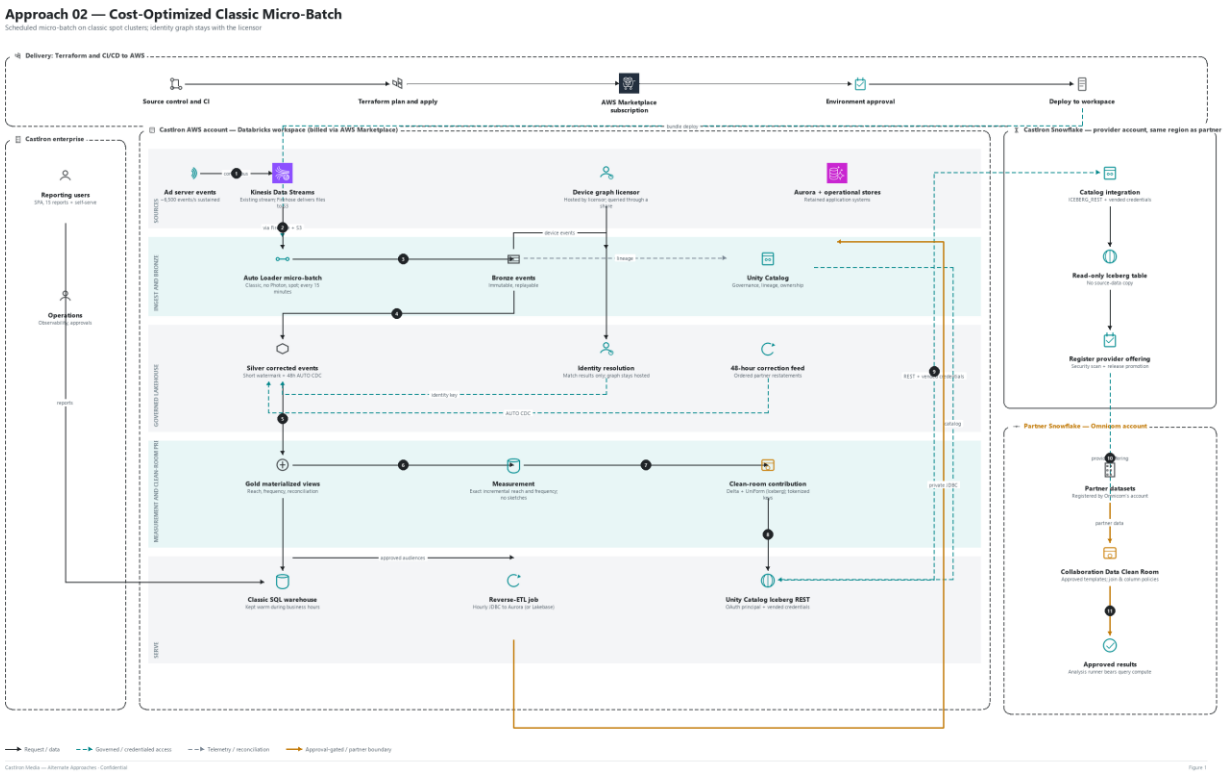


Figure 1. Approach 02 architecture — scheduled micro-batch, hosted identity

2.1 How this approach delivers Phase 1

Phase 1 requirement	How this approach delivers it
Streaming ingest	Existing Firehose delivers files to S3; Auto Loader processes them every 15 minutes with Structured Streaming jobs (bronze, silver, gold). Same stored-procedure decomposition and shadow/parity process as the recommended approach; quality checks built in code.
Identity resolution	Graph remains with the licensor and is queried through a share; only match results are stored.
Measurement	Exact incremental reach and frequency from the household-day aggregate; no sketches.
Clean room	Same pattern as the recommended approach: a CastIron-owned, same-region Snowflake provider account links a read-only Iceberg offering into Omnicom's Collaboration Data Clean Room — no row-level reads from the partner's account.
Reverse-ETL	Hourly Spark JDBC upserts into Aurora MySQL.
Reporting and orchestration	Classic SQL warehouse, kept warm during business hours; Lakeflow Jobs orchestration.

2.2 How it performs

Metric	Expected performance
Event to silver latency	15 to 30 minutes (Firehose buffer + 15-minute schedule)
Event to report marts (gold)	30 to 45 minutes

Burst handling	Backlog builds if a batch exceeds its interval; manual resizing
Identity daily match	~1 to 2 hours; depends on the licensor's data layout and region
30-day reach and frequency refresh	~2 to 3 hours daily
SPA packaged reports	p95 3 to 8 s; 4 to 6 minute cold start if the warehouse is stopped
Self-serve builder	Several seconds on large slices
Reverse-ETL freshness	Hourly
Recovery	Re-run failed batch; more manual tuning

2.3 Approximate monthly cost by resource and workload

Workload	Resource and sizing	Usage basis	Databricks (USD/mo)	AWS (USD/mo)	Total (USD/mo)
Event ingest	Existing Firehose with S3 delivery (billed per 5 KB record increment)	24x7	-	370 to 2,450	370 to 2,450
Streaming micro-batch	Jobs compute, no Photon; driver + 4 x r6gd.2xlarge (spot), instance pool	~12 effective h/day	550	530	1,080
Late data and 48 h restatement	Jobs compute; driver + 4 x r6gd.xlarge (spot)	~2 h/day	50	40	90
Identity resolution (hosted)	Jobs compute, no Photon; driver + 8 x r6gd.2xlarge (spot)	~1.5 h/day	120	80	200
Measurement	Jobs compute, no Photon; driver + 4 x r6gd.2xlarge (spot)	~5 h/day	230	180	410
Clean-room contribution	Jobs compute; driver + 2 x r6gd.xlarge; partner read requests	~1 h/day	15	35	50
Reverse-ETL	Jobs compute from instance pool	Hourly, ~5 min/run	30	30	60
SPA reporting	Classic SQL warehouse, Small (12 DBU/h) + EC2 in own account	Kept warm ~16 h/day	1,270	1,500	2,770
Lakehouse storage	S3 Standard (no resident graph)	~23 TB avg + requests	-	610	610
Network and security	NAT (2 AZ), VPC endpoints, KMS, CloudWatch	24x7	-	300	300
Non-production	Dev + staging (~15% of production compute)	Auto-terminate	500	200	700
Total (steady state)			**2,765**	**3,875 to 5,955**	**~6,640 to 8,720**

Annual run cost: approximately USD 80,000 to 105,000. The Firehose range depends on whether events are aggregated into larger records (Firehose rounds each record up to 5 KB). Firehose is already part of today's bill. Licensor hosting fees for the identity graph are not included, and this excludes the separately billed Snowflake clean-room account (~USD 6,000 to 14,400/year, see the Technical Approach document Section 6).

2.4 Drawbacks

- Data latency of 30 to 45 minutes; not suitable for near-real-time reporting.
- Most engineering and operations effort of the three approaches: custom quality, recovery and tuning; more incidents to manage.
- Retains the Firehose dependency; Firehose cost can reach ~USD 2,450/month after Redshift retirement if records are not aggregated.
- Hosted identity limits join optimization compared with a resident graph.
- Classic SQL warehouse cold starts and slower interactive performance for the SPA.

3. Approach 03: Fully Serverless, Performance-Optimized

All workloads run on Databricks serverless compute: continuous serverless pipelines, serverless jobs and a larger serverless SQL warehouse. There are no clusters to size or patch, and capacity scales automatically.

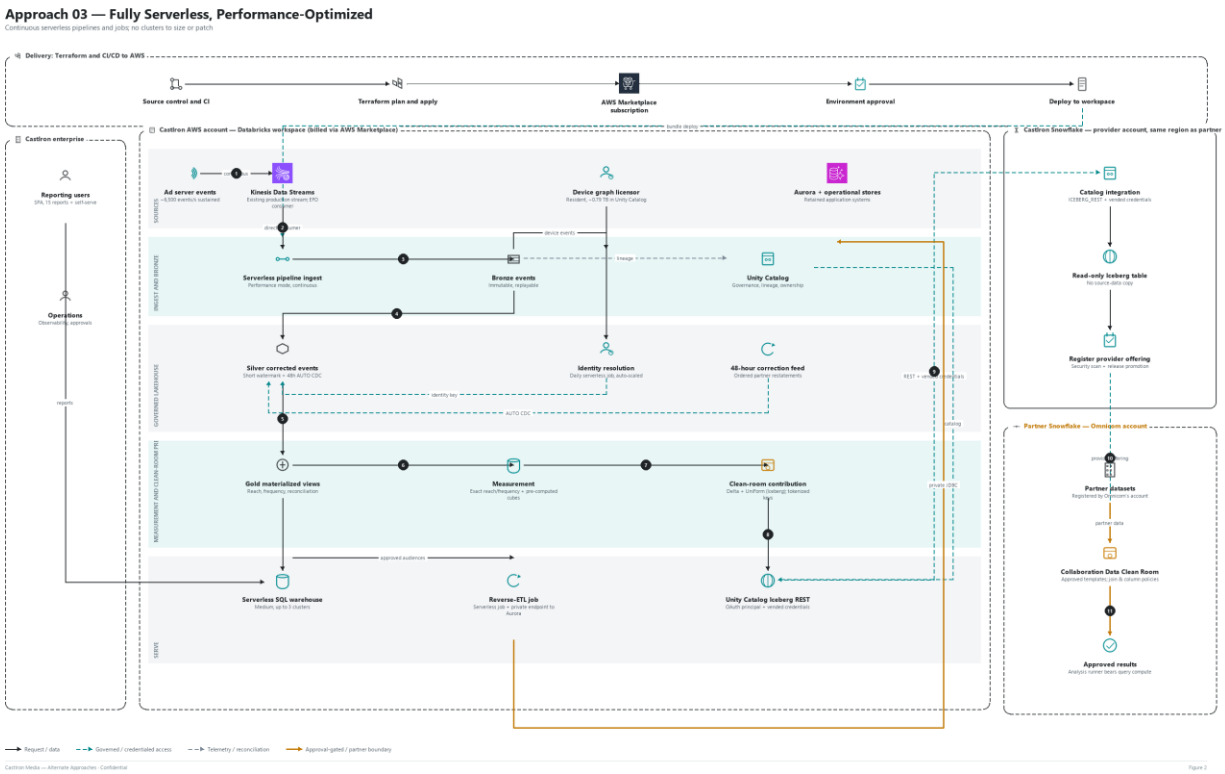


Figure 2. Approach 03 architecture — continuous serverless, resident identity

3.1 How this approach delivers Phase 1

Phase 1 requirement	How this approach delivers it
Streaming ingest	Kinesis EFO consumer into serverless declarative pipelines in performance mode (continuous). Same stored-procedure decomposition and shadow/parity process as the recommended approach.
Identity resolution	Resident licensed graph; daily serverless job with automatic scaling.
Measurement	Exact reach and frequency plus pre-computed cubes for the most-used report dimensions.

Clean room	Same pattern as the recommended approach: a CastIron-owned, same-region Snowflake provider account links a read-only Iceberg offering into Omnicom's Collaboration Data Clean Room — no row-level reads from the partner's account.
Reverse-ETL	Serverless jobs writing to Aurora through a private endpoint (network connectivity configuration).
Reporting and orchestration	Serverless SQL warehouse, Medium, up to 3 clusters for high concurrency; Lakeflow Jobs orchestration.

3.2 How it performs

Metric	Expected performance
Event to silver latency	Under 1 minute
Event to report marts (gold)	2 to 5 minutes
Burst handling	Elastic, no capacity planning
Identity daily match	~15 to 25 minutes
30-day reach and frequency refresh	Under 1 hour
SPA packaged reports	p95 1 to 3 s; instant start; high concurrency
Self-serve builder	Sub-second
Reverse-ETL freshness	15 minutes to hourly
Recovery	Fully managed retries and restarts

3.3 Approximate monthly cost by resource and workload

Workload	Resource and sizing	Usage basis	Databricks (USD/mo)	AWS (USD/mo)	Total (USD/mo)
Event ingest	Kinesis EFO consumer, 12 shards	24x7; ~12.6 TB read	-	300	300
Streaming pipeline	Serverless declarative pipelines, performance mode	24x7	5,600	-	5,600
Late data and 48 h restatement	Serverless jobs	~2 h/day	150	-	150
Identity resolution	Serverless jobs; graph on S3 (0.79 TB)	~25 min/day	280	20	300
Measurement	Serverless jobs incl. pre-computed cubes	~1.5 h/day	600	-	600
Clean-room contribution	Serverless jobs; partner read requests	~1 h/day	50	20	70
Reverse-ETL	Serverless jobs + private endpoint (load balancer) to Aurora	Every 15 to 60 min	100	50	150
SPA reporting	Serverless SQL warehouse, Medium (24 DBU/h), up to 3 clusters	~12 active h/day	6,050	-	6,050
Lakehouse storage	S3 Standard (Delta bronze, silver, gold)	~23 TB avg + requests	-	630	630

Network and security	Private endpoints, reduced NAT, KMS, CloudWatch	24x7	-	300	300
Non-production	Dev + staging (serverless)	On demand	1,400	100	1,500
Total (steady state)			**14,230**	**1,420**	**~15,650**

Annual run cost: approximately USD 188,000. Serverless streaming is estimated at about 1.4 to 1.5 times the equivalent classic cost, because serverless rates include the compute infrastructure. This excludes the separately billed Snowflake clean-room account (~USD 6,000 to 14,400/year, see the Technical Approach document Section 6).

3.4 Drawbacks

- Highest run cost of the three approaches: about 1.6 times the recommended approach, driven by continuous serverless streaming and the larger SQL warehouse.
- Premium unit pricing is paid on a steady 24/7 load where classic compute is cheaper.
- Less control over instance types and tuning; EC2 Savings Plans and spot pricing cannot be applied.
- Private access to Aurora and other VPC resources requires additional network connectivity configuration.

4. Comparison with the Recommended Approach

Criterion	Approach 01: Recommended	Approach 02: Cost-Optimized	Approach 03: Fully Serverless
Event to report latency	5 to 10 min	30 to 45 min	2 to 5 min
SPA report performance	p95 2 to 5 s	p95 3 to 8 s, cold starts	p95 1 to 3 s
Monthly run cost	~USD 9,850	~USD 6,640 to 8,720	~USD 15,650
Annual run cost	~USD 118,000	~USD 80,000 to 105,000	~USD 188,000
Operational effort	Moderate	High	Low
Main drawback	Two compute models to govern	Latency and engineering effort	Highest run cost

Recommendation unchanged

Approach 01 remains recommended (see the Technical Approach document). Its annual run cost (~USD 118,000) is higher than Approach 02's (~USD 80,000 to 105,000) but about 37% lower than Approach 03's (~USD 188,000), while delivering materially better latency and SPA performance than Approach 02. Approach 02 is the right call where a 30 to 45 minute reporting delay is genuinely acceptable and minimizing run cost matters more than engineering effort. Approach 03 is the right call where operational simplicity is worth paying a premium for and the 24/7 serverless rate is acceptable.